Chapter 1 - Fast-Moving Consumer Goods

In the first chapter, you will be working as a category manager for Healthmax, a FMCG company active in the shampoo business. You will calculate Key Performance Indicators (KPIs) such as market share and value growth on various time dimensions such as year-to-date (YTD) and moving annual total (MAT).

—

Investigating the dataset

You are the Category Manager for the company HealthMax, market leader in the shampoo industry. You are being asked to analyze the available market and to come with actionable insights to grow the business.

The first step of every analysis is to do a sense-check of the dataset.

—

Note that every line contains the unit and value sales for a `Brand`, sold in a specific `Region`, during a specific `Month` and `Year`. Every brand belongs to only one `Subcategory` and only one `Supplier`.

Apparently there is a hardcoded dollar sign ($) in the monthly sales values, which are in the `Values Month` column.

- Remove the dollar signs.
- Reformat the column into currency formatting.

—

- Create a PivotTable to make an overview of the different `Brand`s per `Supplier`.
- Rename the new worksheet to "Brands per Supplier".
- Select the `External_Data` table.
- Navigate to *Insert* > *PivotTable*.
- Click *OK* to create the PivotTable in a new worksheet.
- Add the right fields to the Rows area of your PivotTable (in that order): `Supplier`, `Brand`.

—

How many brands on the market are owned by HealthMax?

2

—

Year-over-year growth

HealthMax's management wants to know how their brands evolved over the last couple of years. Your job is to calculate the year-over-year growth per brand.

If you lost progress, close any open reports and load `1_2_growth_calculations.xlsx` from the Workbooks folder.

—

- Create a PivotTable in a new Worksheet.
- Rename the new Worksheet to "HealthMax Growth".
- Display the value sales per brand for all years.
- Filter on HealthMax as a supplier.

—
Change the displayed values to growth numbers compared to the previous year by right-clicking the values in the PivotTable and using the *Show Values As* section.
—
We only have 3 months of data for 2023, so it doesn't make sense to compare it to 2022. Hide 2023 in your pivot table.
—
In which year did HealthMax realize its strongest growth?
2020
Well done, in 2020 HealthMax grew by 4%! Shinez contributed to this success with its 7,7% YoY growth.

Year-to-Date

The dataset contains unit and value sales per month for all years from 2018 to 2023. You would however like to look at different time periods to calculate your KPI's. Let's calculate the Year-To-Date (YTD) for all the dataset points in this exercise.

If you lost progress, close any open reports and load `1_3_year-to-date.xlsx` from the Workbooks folder.

—
Go back to the `External Data` tab and add a new column to the table, call it "Units YTD".
—

Let's first calculate the yearly total on the most granular level: unit sales by brand by region by year. We'll adapt the formula to change the calculation to YTD in the next step.

You can calculate the YTD numbers by using a `SUMIFS()` function:

- Use `Units Month` as a sum range.
- Add the column `Brand` as the first *criteria_range* and then the first cell in the column `Brand` as the criterium.
- Repeat the previous step for the columns `Region` and `Year`.

—
Add `Month` as the last *criteria_range* to filter for YTD numbers to the `SUMIFS()` you just created. Your final criterium should look like this: `"<="&[@Month]`.
—

- Create a new column in your table and calculate the `Values YTD` using again the `SUMIFS()` formula.
- Reformat the column into currency formatting.

—
Manually verify in the first rows of your dataset whether your calculated `Units YTD` and `Values YTD` are correct.

- Select some `Units Month` cells that belong to the same `Brand`, `Region` and `Year`
- Check if the sum of those cells corresponds to the calculated `Units YTD` in the row which contains the latest `Month` in your selection.
- Repeat these steps to verify the `Values YTD`

—
What are the YTD value sales for RedRose in 07/2018 for the central region?
$452940

Moving Annual Total

You calculated the YTD values in the previous exercise but would also like to look at a longer period than YTD. Let's calculate the Moving Annual Total (MAT) for all the dataset points in this exercise.

Reminder: MAT is the sum of the latest 12 months.

Moving Annual Total (MAT)												
Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Year -1	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Year -2	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec

If you lost progress, close any open reports and load `1_4_moving_annual_total.xlsx` from the Workbooks folder.

—
Go back to the `External Data` tab and add a new column to the table, call it "Units MAT".
—

Calculate the `Units MAT` by using the calculated `Units YTD` and adding the remaining values with another `SUMIFS()` function.

Your new `SUMIFS()` function should sum up the corresponding values for the months of the previous year that are still included in your current MAT.

```
Units MAT = ____ +
SUMIFS(____,____,____,____,____,F:F,[@Year]-1,G:G,">"&____)
```

MAT = Moving Annual Total = the sum of the values of the previous 12 months.

```
Units MAT = [@[Units
YTD]]+SUMIFS(H:H,D:D,[@Brand],E:E,[@Region],F:F,[@Year]-1,G:G,">"&[@Month]
).
```

—
Create a new column in your table and calculate the `Values MAT`.
—
Manually verify in the first rows of your dataset if your calculated `Units MAT` and `Values MAT` are correct.
—

- Calculate the total turnover (revenue) of the total shampoo category using a PivotTable using the MAT values you calculated. Make sure you only show the latest MAT values (March 2023).

- Rename the new worksheet to "MAT Value Total Category"

In your PivotTable, you should have:

- Filter: `Year`, `Month`
- Values: `Values MAT`

Make sure you filter on the latest available period (March 2023). If you do not filter you will add up multiple MAT values which won't be correct.

—

What is the total turnover of the total shampoo category in the latest 12 months?
98452906
Great, that's right! In the most recent 12 months, the shampoo's generated a bit over 96 million dollars of turnover.

Market Share

You just calculated the size of the total annual category, let's now try to identify the market share of the biggest players on the market. We'd like to create a graph to see how our brands evolved over the years and see if there are regional differences.

If you lost progress, close any open reports and load `1_5_market_share.xlsx` from the Workbooks folder.

—
Create a new PivotTable to calculate the total value sales per year, per brand in a new worksheet called "Market Share". As we would like to create a graph that shows an evolution over time, put your time-related data in the Rows field.

Your PivotTable should look like the below:

- Columns: `Brand`
- Rows: `Year`
- Values: `Values Month`

—

- Change the displayed values in your PivotTable from the total sales to the market shares per year.
- Sort the brands from big to small.
- Delete the Grand Total for the columns, as we cannot use this calculation in this exercise.

When clicking right on the values you will be able to select *% of Row Total* in the *Show Values As* section, considering you put your `Brand` in the Columns area.

Look at the *More Sort Options* when you click on the little arrow next to *Column Labels* in your PivotTable. Make sure you filter the values and not just the brand names alphabetically.

—
Visualize the data by creating a line chart out of your PivotTable.
—

- Resize your chart to better see the difference between the different top suppliers.
- Put the legend at the bottom of the chart.
- Add the title "Shampoo Brands Market Share Value Evolution" above the chart.
- Hide all field values on the chart.

—
Add a slicer to see if the market share evolution of the brands is different in different regions of the country.

You can insert a slicer `Region` to the PivotTable or PivotChart to dynamically change the table or graph.

- Select the PivotTable or the PivotChart.
- Navigate to *PivotTable Analyze* or *PivotChart Analyze* > *Filter* > *Insert Slicer*.
- Select `Region` and click on *OK*.

—

HealthMax owns Starbust and Shinez. In which region does Starbust have the biggest market share in 2023?

Center

Correct! HealthMax has 21.59% of market share in the Center and is performing better in the Center than in the rest of the country!

New strategy implementation

Your biggest competitor, FreshCo Industries, decided to stop their TV advertisements as from January 2023 and shift all investments to digital advertising. In March 2023 you see that:

- Their monthly market share is 26%
- Their YTD market share is 28%
- Their MAT market share is 30%

Would you, assuming that everything else stayed the same, recommend your management to also switch all investments from TV advertising to digital advertising? Tip: understanding how the percentages change over time will help you solve this problem.

No, the new strategy is not working well for FreshCo Industries.

Perfect! MAT values look at values of the last 12 months, or from April 2022 to March 2023. Their market share over that period is 30%, but both YTD and monthly values are lower: 28% and 26%, meaning their market share is declining. It is therefore not a good idea to shift all investments to digital advertising.

Your recent learnings

When you left 22 hours ago, you worked on Fast Moving Consumer Goods, chapter 1 of the course Case Study: Net Revenue Management in Excel. Here is what you covered in your last lesson:

You learned about the Fast Moving Consumer Goods (FMCG) industry and its characteristics. FMCG products are everyday items like food, beverages, and personal care products that are frequently purchased, consumed quickly, and sold in large volumes at low margins.

Key points covered include:

- **FMCG Characteristics**:
  - Frequently purchased and consumed quickly
  - Relatively low priced and widely available
  - High shelf rotation due to high demand or short expiration dates
- **Key Performance Indicators (KPIs)**:
  - **Market Share**: Measures a brand's sales as a percentage of total market sales.

Growth: Assesses how much a brand or company is growing over time using the formula:
```
Growth = (Current Period Sales / Previous Period Sales) - 1
```

- **Time Periods for KPIs**:
  - **Year-to-Date (YTD)**: Sales from the beginning of the year to the current month.
  - **Moving Annual Total (MAT)**: Sales over the last 12 months.
- **Excel Formulas for KPIs**:

YTD Calculation:
```
=SUMIFS(H:H, D:D, [@Brand], E:E, [@Region], F:F, [@Year], G:G, "<=" & [@Month])
```
You also explored how to analyze datasets to calculate these KPIs and interpret market trends.

The goal of the next lesson is to explore advanced strategies for optimizing product pricing and portfolio management to maximize profitability.

Chapter 2 - Net Revenue Management

In this chapter, you will learn about the concept of Net Revenue Management (NRM) and will focus on two of its pillars: Brand Portfolio Pricing and Mix Management. You'll calculate important KPIs such as net sales contribution and gross margin. You'll also use brand Portfolio Pricing to identify the correct price for your consumers. Consecutively you'll use Mix Management techniques to help Healthmax launch a new product and expand into a new subcategory.

—

Net sales

HealthMax is looking to shift investments in between its products and is asking you to make a list of product prioritizations in function of their net sales and their profitability. Let's calculate the net sales for each product in this exercise.

If you lost progress, close any open reports and load `2_1_net_sales.xlsx` from the Workbooks folder.

—

- Apply currency formatting to columns `Retail Price`, `Net Price` and `COGS`.
- Add a thousands separator to `Volume 2022`.

—

Add a row total to the table and make sure the sum of `Volume 2022` is displayed in the total row of your table.
Check the *Total Row* checkbox in the *Table Design* tab.

—

- Create an extra column called "Net Sales 2022" and calculate the total net sales per product.
- Make sure the sum of `Net Sales 2022`is displayed in the total row of your table.
- Apply currency formating to `Net Sales 2022`.

`Net Sales 2022` = `Net Price` x `Volume 2022`.

—

What were the total Net Sales for Healthmax in 2022?
$20480714.90

Gross margin

HealthMax is looking to shift investments in between its products and is asking you to make a list of product prioritization in function of their net sales and their profitability. Let's calculate the profitability for each product in this exercise.

If you lost progress, close any open reports and load `2_2_gross_margin.xlsx` from the Workbooks folder.

—

The table in the `Internal Sales Data` worksheet contains the different prices, costs and volumes for all different products in HealthMax' portfolio.

- Create an extra column called "Gross Profit per unit" and calculate HealthMax's gross profit per unit for all products.
- Apply currency formatting to `Gross Profit per unit`.

`Gross Profit per unit` = `Net Price` - `COGS`.
—

- Create an extra column called "Gross Profit per product" and calculate the total gross profit every product generates.
- Make sure the sum of `Gross Profit per product`in the total row.
- Apply currency formating to Gross Profit per product`.

`Gross Profit per product` = `Gross Profit per unit` x `Volume 2022`.
—

- Add an extra column called "Gross Margin" and calculate HealthMax's gross margin for all products.
- Change the number formatting of your new column to percentage.

`Gross Margin` = `Gross Profit per unit` / `Net Price`.
—

- Calculate your (weighted) average gross margin in the row total.
- Change the number formatting to percentage.

Total `Gross Margin` = Total `Gross Profit per product` / Total `Net Sales 2022`.
—
What is the gross margin for the shampoo's in our portfolio? (%)
68.49%
Well done! 68.5% of the average net selling price is gross profit!

Profitability matrix

HealthMax is looking to shift investments in between its products and is asking you to make a list of product prioritization in function of their net sales and their profitability.

You decide to make a Scatter Plot to compare the gross margin to the net sales contribution for the different products.

If you lost progress, close any open reports and load `2_3_profitability_matrix.xlsx` from the Workbooks folder.

—

- Add a new column to your table called "Net Sales Contribution". Calculate it by dividing the `Net Sales 2022` by the total of `Net Sales 2022`.
- Change number formatting to percentage.

Divide the `Net Sales 2022` by the total shown at the bottom of `Net Sales 2022`.
—

Create a Pivot Table in a new worksheet called "Profitability Matrix".

- Your PivotTable should look like the below:
    - Columns: `Gross Margin`
    - Rows: `ProductID`
    - Values: `Net Sales Contribution`
- You can delete the row total and the column total.
- Change the `Net Sales Contribution` number formatting to percentage.

Your PivotTable should look like the below:

- Columns: `Gross Margin`
- Rows: `ProductID`
- Values: `Net Sales Contribution` (summed)

—

Excel doesn't allow you to create a Scatter Plot straight out of the PivotTable.

- Copy and paste the data right underneath the PivotTable and create a Scatter Plot.
- Switch the rows and columns of the Scatter Plot to have the `Gross Margin` on the horizontal axis and `Net Sales Contribution` on the vertical axis.
- Resize the graph to be a square.
- To insert a Scatter Plot, select the data and navigate to *Charts* under the *Insert* tab.
- Select the Scatter Plot and click on *Switch Row/Column* in the *Chart Design* tab.

—

- Rename the chart to "HealthMax Profitability Matrix".
- Add a Primary Horizontal Axis Title "Gross Margin".
- Add a Primary Vertical Axis Title "Net Sales Contribution".

Select the graph and navigate to *Add Chart Element* under the *Chart Design* tab.
—

Which statement is NOT correct?

The Starbust Strong Hair range is the most important range of the assortment.
Indeed, this statement is not correct! The Strong Hair range is situated at the bottom left corner of the scatter plot, meaning relatively low profitability and net sales.

New category opportunity

When looking at your brand mix, you realize that HealthMax is only active in 2 out of the 5 shampoo subcategories and that it might be interesting to launch a new product to enter a new subcategory. To do so, you will identify the fastest growing subcategory (in units) and make an estimation of the total subcategory size for 2024.

If you lost progress, close any open reports and load `2_4_new_category_opportunity.xlsx` from the Workbooks folder.

—

To identify the fastest growing category, let's go back to the `External Data` worksheet.

- Create a new pivot table to identify the subcategory that realized the biggest growth when comparing 2022 to 2018.
- Make sure you calculate the growth in units and not in values.
- Call the new worksheet "New Category Opportunity".

In your PivotTable, you should look like the below:

- Rows: `Subcategory`.
- Columns: `Year`.
- Values: `Units Month`.

In your PivotTable:

- Right click on the values and select *% Difference From…* in the *Show Values As* section. For the Mac users: go to *More Options…* of the *Show Values As* section.
- Make sure you compare the values to the year 2018.
- You can delete the Grand Totals.

—

- Filter your years and only keep 2018 and 2022.
- Have a look at the displayed values, to determine the fastest growing subcategory (in units) since 2018.

Click on the filter button right next to *Column Labels*.
Organic is the fastest growing subcategory at 275.88%.
—

Now that we have determined the fastest growing subcategory, we would like to make a full year estimate of the size (in units) of this subcategory for 2024. Let's see what the latest yearly size of this subcategory was, using the `Units MAT` values for March 2023 (latest data available).

- Create a new PivotTable right underneath to display the `Units MAT`, for the fastest growing subcategory. Do not forget to filter for the latest datapoint (March 2023).

In your PivotTable, you should look like the below:

- Rows: `subcategory`.
- Columns: `Year` and `Month`.
- Values: `Units MAT`.

Filter the row values to the fastest growing subcategory and apply the right filters to only show the March 2023 `Units MAT`.

—

Now that you calculated the latest size of that subcategory, you would like to make an estimation of the size for the full year 2024. Looking at the historic trend, you estimate that this category size for the full year 2024 will be 20% higher than the March 2023 MAT values.

Make an estimation of the total units sold in 2024 in your identified subcategory and store it in a cell underneath your PivotTable with a short description of the figure on its left.

Multiply the `Units MAT` you calculated in the previous step by (1+20%).
—
How many shampoos will there be sold in 2024 in the fastest growing shampoo subcategory?
1,020,877
Very good! The organic shampoos are clearly booming and there is a big potential for HealthMax, as market leader, to recruit a whole new customer base in this segment!

New product launch

Now that you discovered that organic shampoos are booming, your company has two possible products that fit in this category and that can be launched in the beginning of 2024. Let's identify the best candidate by making an estimation of expected net sales, gross profit, and gross margin for both products.

If you lost progress, close any open reports and load `2_5_new_product_launch.xlsx` from the Workbooks folder.

—

Marketing sent you a table with the given data for both products.

- Import `new_product_launch.csv` from the `Datasets` folder.
- Rename the worksheet to "Organic Shampoo Launch".
- Apply percentage number formatting to `Estimated Unit Market Share`.
- Apply currency number formatting to `Net Price` and `COGS`.

To import a CSV file go to *Data* > *From Text/CSV*.
—

- Calculate the column `Estimated units sold 2024`, based on the market size you calculated in your previous exercise
- Apply a thousands separator to your values.

Your formula should look like this `New Category Opportunity'!$B$18*[@[Estimated Unit Market Share]]`.

Note that you might change the cell `B18`, depending on where you stored the result of your previous exercise.

—

- Make an estimation of the Net Sales per product.
- Make sure your column has currency number formatting.

`Net Sales` = `Estimated units sold 2024` x `Net Price`.
—

- Calculate the `Gross Profit per unit` and `Gross Profit per product` for both products.
- Make sure your column has correct number formatting.

`Gross Profit per unit` = `Net Price` - `COGS`.

`Gross Profit per product` = `Estimated units sold 2024` x `Gross Profit per unit`

Gross Profit is always expressed in dollars.

—

- Calculate the `Gross Margin` for both products.
- Apply the correct number formatting.

`Gross Margin` = `Gross Profit per unit` / `Net Price`.

Gross Margin is always expressed in %.

—

The goal of Healthmax is to increase their gross profit as much as possible. Which product should they launch next year?

They should launch Herbashine, it will generate 0.34 M$ in net sales.
Correct! Herbashine will generate less net sales than HerbEssentials, but more gross profit!

Your recent learnings

When you left 4 days ago, you worked on Net Revenue Management, chapter 2 of the course Case Study: Net Revenue Management in Excel. Here is what you covered in your last lesson:

You learned about the concept of Net Revenue Management (NRM) and its importance in driving growth for FMCG companies. Specifically, you focused on two pillars: Brand Portfolio Pricing and Mix Management. Here's a recap of the key points:

- **Net Sales and Gross Margin**: You calculated net sales by subtracting discounts from total revenue. Gross margin was determined by the ratio of gross profit (net sales minus COGS) to net sales.
- **Brand Portfolio Pricing**: This involves setting the right price for your products based on consumer willingness to pay and competitor pricing. For instance, if your cookies are superior, they might justify a premium price.
- **Mix Management**: This strategy focuses on optimizing your product portfolio for profitability. You learned to prioritize products not just by sales volume but also by their gross margin.

You also engaged in practical exercises to reinforce these concepts:

Net Sales Calculation:

```
Net_Sales_2022 = Net_Price * Volume_2022
```

-

Gross Margin Calculation:

```
Gross_Profit_per_unit = Net_Price - COGS
Gross_Profit_per_product = Gross_Profit_per_unit * Volume_2022
Gross_Margin = Gross_Profit_per_unit / Net_Price
```

-

These exercises helped you understand how to analyze and prioritize products based on their financial performance.

The goal of the next lesson is to apply the principles of Trade Terms Management, Pack Price Architecture, Promotion Management, and Forecasting to real-world business scenarios.

Chapter 3 - Optimizing Net Revenue

In the last chapter, you will focus on the three remaining pillars of Net Revenue Management (NRM). Trade Terms Management will help you to decide in which customers and products you should be investing to grow your business. You'll combine that information with Pack Price Architecture to develop a new pricing strategy for a small shampoo bottle and use a Promotion Management technique to identify the most successful promotion. Finally, you'll wrap it up by forecasting next year's growth by combining organic growth with the additional revenue of your NRM activities.
—

Price Pack Architecture

You see that the smallest shampoo size in your portfolio is 100ml. Research has shown that people that travel are looking for shampoo bottles of 50ml. You contact your colleagues in other countries that do have a 50ml shampoo bottle in their assortment. They told you the small format is very successful and shared some assumptions so you can build your business case. Let's calculate the potential of a smaller shampoo bottle.

If you lost progress, close any open reports and load `3_1_price_pack_architecture.xlsx` from the Workbooks folder.

—

- Make a duplicate of the `Internal Sales Data` tab and call it "50ml Shampoo 2024".
- Delete the total row and the column `Net Sales Contribution`.

Right-click on the tab you would like to duplicate and select *Move or Copy*.
—

Let's look at the pricing of our current products.

Use the `Retail Price` to calculate the `Price per ml`, with 3 decimals, in a new column.

`Price per ml` = `Retail Price` / `Pack Size (ml)`
—

You decide to launch your best rotating product, the Starbust Ultra Soft 100ml, in a 50ml format.

- Add a row to your table and fill in the first four columns.
- To define the retail price, you can take a 50% mark-up of the `Price per ml` of the 100ml pack.

Your formula should look like this: `Retail Price`= `Pack Size (50 ml)` * Price per ml of the 100ml pack * (1+50%).
—

- The `Net Price` is $2.30 and the `COGS` amount to $0.70.

- Fill in the remaining columns in your table and assume that the `Volume 2022` will be 10% of the volume of the 100ml pack in 2022.

The `Volume 2022` of the 50ml pack should be 10% x `Volume 2022` of the 100ml pack.
—
How much yearly net sales (in thousands) would the 50ml format generate?
$265960
Well done! The 50ml format will target a new consumer segment and will generate 265,960 dollars of net sales!

Promotion Management

HealthMax played three different promotions on the Shinez brand in 2022. To build the strategy for 2024, you'd like to come with clear recommendations of which promotions are the most effective ones.

Let's calculate the ROI for the three different promotions.

If you lost progress, close any open reports and load `3_2_promotion_management.xlsx` from the Workbooks folder.

—

Let's go back to the `External Data` tab to analyze market data.

- Create a pivot table with the value sales by month for the brand "Shinez" in 2022 in a new worksheet called "Promotion Graph".

Your PivotTable should look like the below:

- Filter: `Brand`, `Year`
- Rows: `Month`
- Values: `Values Month`

Make sure you apply the correct filters.

—

- Create a PivotChart out of the PivotTable to visualize the sales numbers.
- Rename the *Chart Title* to "Shinez Value Sales 2022".

Select the PivotTable and navigate to *Insert* > *Charts* > *PivotChart* > *Line Chart*.

If you are working on a Mac, you can change the Chart Type in the *Design* tab.

—

- Import `promotion_analysis.csv` from the `Datasets` folder.
- Rename the worksheet to "Promotion Analysis"

To import a CSV file go to *Data* > *From Text/CSV*.
—

- Fill in the `Value sales` by using `VLOOKUP()`. Look up the `Values Month` in the PivotTable in `Promotion Graph` for the corresponding month in the table.
- Calculate the `Baseline Sales` by taking the average `Values Month` of the months where no promotion was active. Don't forget to lock your cells in the formula.
- Apply currency formatting to both columns.

- `Value sales` = `VLOOKUP([@Month],PivotTable in Promotion Graph,Month array in PivotTable,FALSE)`. Use *F4* to lock your cells in your formula.
- `Baseline sales` = `=AVERAGE('Promotion Graph'!$C$5:$C$7,'Promotion Graph'!$C$9:$C$11,'Promotion Graph'!$C$13:$C$14,'Promotion Graph'!$C$16)`. Use the *ctrl button* to select multiple values. Use *F4* to lock your cells in your formula.

—

Calculate the generated `Uplift` per promotion.

`Uplift` = `Value sales`-`Baseline sales`.

—

Calculate the `ROI` per promotion.

ROI = (`Uplift` - `Cost`) / `Cost`.

—

Which promotion is the most effective one?

Promotion #2: Buy 2, get 20% off

Correct! The ROI for this promotion is 12% and is higher than Promotion #3. The ROI of Promotion #1 is negative and should not be repeated.

Forecasting

Now that you calculated different opportunities based on the NRM pillars, it is time to wrap up and present the numbers to your management. You would like to present a waterfall graph to show how your initiatives will have an impact on the company's net sales of 2024.

Let's start by making a forecast for the net sales in 2024 without our NRM initiatives. We'll base ourselves on the market sales until 2022, extrapolate the current trend until 2024 and calculate what it represents in net sales. You will create the waterfall graph in the next exercise.

If you lost progress, close any open reports and load `3_3_forecasting.xlsx` from the Workbooks folder.

—

Let's start by forecasting the estimated total market sales of HealthMax for 2023 and 2024.

- Create a PivotTable which shows the yearly value sales of HealthMax until 2022.
- Rename the worksheet to "Sales until 2022".
- Remove the Grand Total in your PivotTable.
- Apply currency formatting to the values in your PivotTable.

Since we are looking for market sales, you should use the `External Data` table for this exercise.

In your PivotTable, you should have:

- Rows: `Year`
- Filter: `Supplier'
- Values: `Values Month`

Don't forget to filter on "HealthMax". In the row filters, you can unselect the `Year` 2023.

—

- Make a forecast for the full year 2023 and 2024 using the *Forecast Sheet* function.
- Rename the header of column B `Values` to `Market values` to avoid confusion.
- Rename the new worksheet to "Forecast 2024".

Since Forecast Sheet is not available on Mac, use =FORECAST.ETS(Target Date,Values,Timeline)

Select the years and values in your PivotTable and navigate to *Data* > *Forecast* and click on *Forecast Sheet*. Make sure your forecast ends in 2024.

—

You forecasted the market values for 2023 and 2024. Let's calculate the amount in Net Sales to see how this will impact the company's revenue.

- Add two columns to the table called "Net Sales" and "Ratio".

- Look up the `Net Sales 2022` in the `Internal Sales Data` tab, add it to the table for the year 2022.
- Calculate the `Ratio` for 2022 by dividing the `Net Sales` by the `Market values`.

The total `Net Sales 2022` amounted up to $20,480,715. `Ratio` = `Net Sales` / `Market values`

–

Calculate the `Net Sales` for 2023 and 2024, using the forecasted values and the `Ratio` of 2022.

Your formula should look like this : `Net Sales` = `Forecast`*`Ratio of 2022`

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What are the currently forecasted net sales for HealthMax in 2024 ?

$20,808,712.14

Very good! It seems like HealthMax's net sales will keep on increasing, based on the latest market trends!

Waterfall

Now that you calculated different opportunities based on the NRM pillars, it is time to wrap up and present the numbers to your management. You would like to present a waterfall graph to show how your initiatives will have an impact on the company's net sales of 2024.

Let's use the forecast you calculated in the previous exercise, add the NRM pillars and make a clear waterfall graph out of it!

If you lost progress, close any open reports and load `3_4_waterfall.xlsx` from the Workbooks folder.

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- Create a new worksheet called "Waterfall"
- Add the following columns to the first row of the worksheet (in that order): `Estimated net sales 2023`, `Natural growth`, `Organic shampoo`, `50ml shampoo`,`Estimated net sales 2024`.

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- Look up the 2023 `Estimated Net Sales 2023` in the `Forecast 2024` sheet so that it shows the value in the `Waterfall` sheet.
- `Natural growth` is the difference between the forecasted `Net sales` of 2024 and 2023. Add it to the `Waterfall` sheet as well.

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Let's not forget to add the extra NRM initiatives you identified!

- Look up the generated `Net sales` of the launch of the `Organic shampoo` (product 2) and the `50ml shampoo`.

You wanted to launch product 2 in the `Organic Shampoo Launch`, make sure you pick the `Net sales` and not the `Gross Profit per product`.

—

Let's assume the growth initiatives don't cannibalize each other.

- Estimate the 2024 sales by adding all initiatives (including the natural growth) together.

`Estimated net sales 2024` = `Estimated net sales 2023` + `Natural growth` + `Organic shampoo` + `50ml shampoo`

–

- Create a waterfall graph from your table and name it "Estimated Net Sales 2024".

- Set `Estimated net sales 2024` as a total.
- To create a waterfall graph, select your table and navigate to *Insert* > *Charts* > *Waterfall*. If you don't find it straight away, you can also find it by clicking on _Recommended Charts > _All Charts_.
- To set `Estimated net sales 2024` as a total: select the bar in the graph and navigate to *Format* > *Format Selection and* tick the box *Set as total*.
    - On Mac, double click the specific bar, right click, select Format Data Point, check the box “Select as total”

—

What are HealthMax' estimated net sales for 2024 with your identified NRM opportunities?

$21,411,569.18

Very good! As you can see, your NRM initiatives generate more net sales than the natural growth. Your management is very pleased to see that you found some great NRM opportunities to generate more net sales and will also follow your recommendations on "Mix Management" and "Promotion Management" to further improve the gross margin!